10.3: Isomorphisms of Graphs

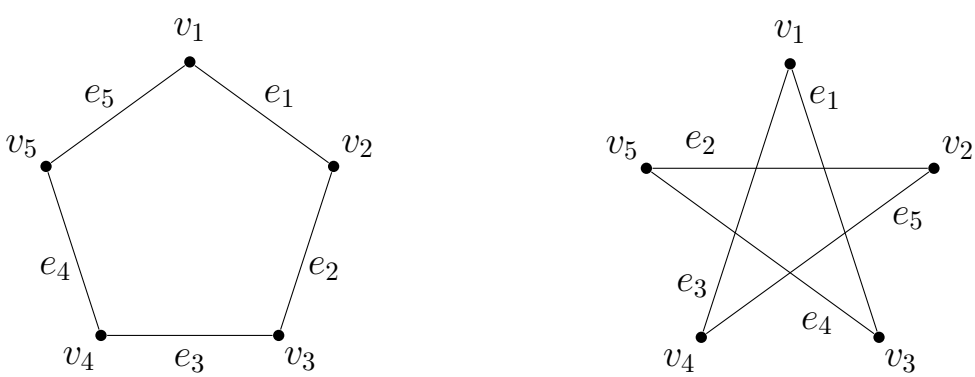
Definition.
Let G and G' be graphs with vertex sets $V(G)$ and $V(G')$ and edge sets $E(G)$ and $E(G')$, respectively. G **is isomorphic to** G' if, and only if, there exist one-to-one correspondences $g : V(G) \to V(G')$ and $h : E(G) \to E(G')$ that preserve the edge-endpoint functions of G and G' in the sense that for each $v \in V(G)$ and $e \in E(G)$,

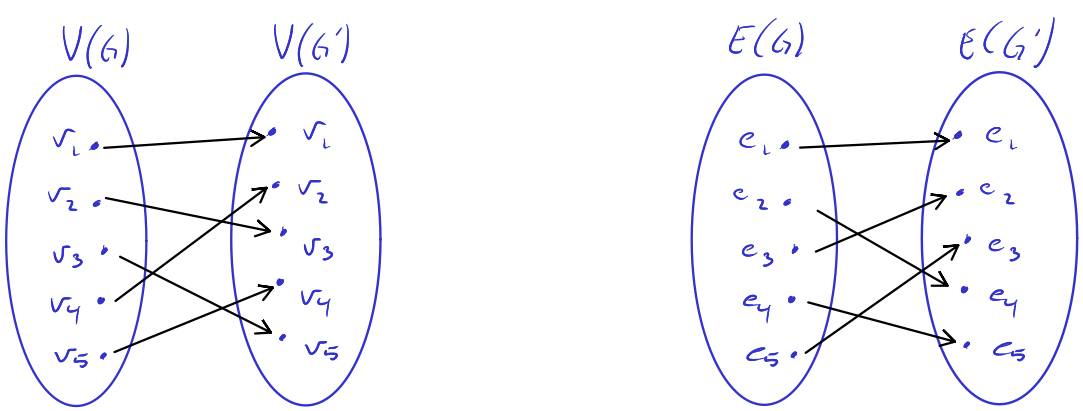
$$v \text{ is an endpoint of } e \Leftrightarrow g(v) \text{ is an endpoint of } h(e).$$

If G and G' are simple (no loops or parallel edges), then G is isomorphic to G' if, and only if, there exists $g : V(G) \to V(G')$ such that

$$\{u, v\} \text{ is an edge in } G \Leftrightarrow \{g(u), g(v)\} \text{ is an edge in } G'.$$

Example. Show that the following graphs are isomorphic:





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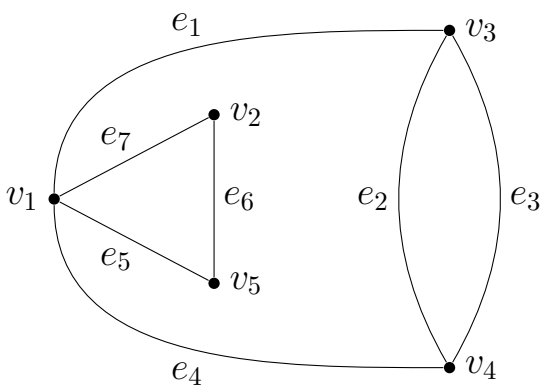
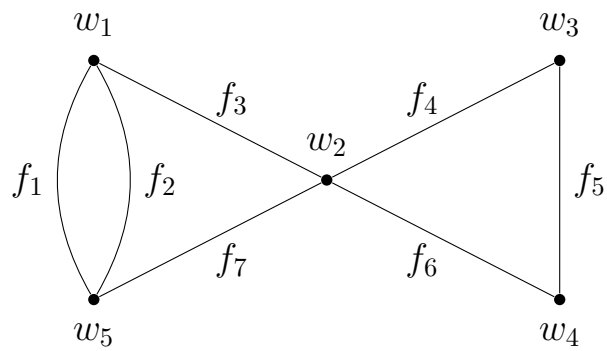
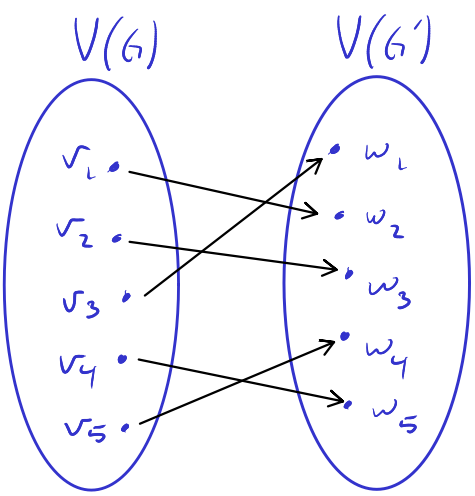
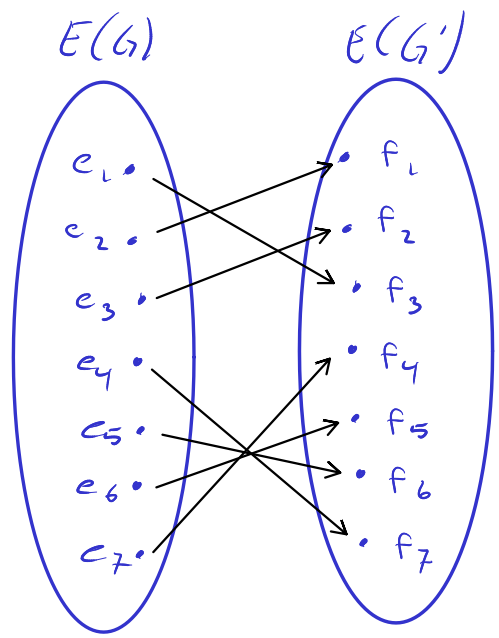








Theorem 10.3.1
Let S be a set of graphs and let R be the relation of graph isomorphisms on S. Then R is an equivalence relation on S.

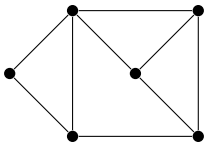
Definition.
A property P is called an **invariant for graph isomorphism** if, and only if, given any graphs G and G', if G has property P and G' is isomorphic to G, then G' has property P.

Theorem 10.3.2
Each of the following properties is an invariant for graph isomorphism, where n, m and k are all nonnegative integer:

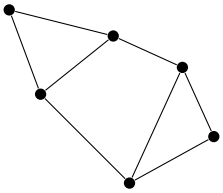
1. has n vertices
2. has m edges
3. has a vertex of degree k
4. has m vertices of degree k
5. has a circuit of length k
6. has a simple circuit of length k
7. has m simple circuits of length k
8. is connected
9. has an Euler circuit
10. has a Hamiltonian circuit.

Example. For the following pairs of graphs, determine if the graphs are isomorphic. Explain why or why not.

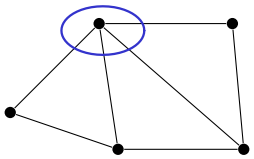
Degree 18

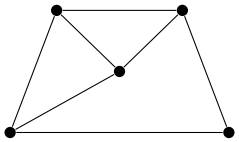


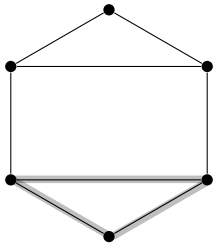
Degree 16



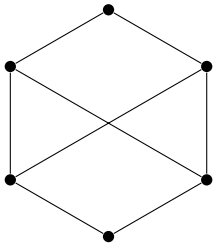
Degree 4







Simple circuit of length 3



No simple circuit of length 3